

SAR Automatic Target Recognition Based on Attribute Scattering Center Model and Discriminative Dictionary Learning

Tingli Li¹ and Lan Du¹, *Senior Member, IEEE*

Abstract—Synthetic aperture radar (SAR) automatic target recognition plays a crucial role in both national defense and civil applications. Although many methods have been proposed, it remains one of the most challenging tasks to identify different terrain vehicle targets. In this paper, a novel method is proposed based on the attribute scattering center model (ASCM) and discriminative dictionary learning. This method contains three main stages. In the first stage, the low-level local features are generated by the convolution operation between SAR patches and the ASCMs with different parameters. These parameters are selected by genetic algorithm in advance with labeled patches. In the feature coding stage, a discriminative dictionary learning method is proposed with the label consistent and locality constraint on the sparse coefficient to incorporate the label information and local geometric information into the sparse representation, which is called label consistent and locality constraint discriminative dictionary learning (LcLcDDL) method. The low-level local features are sparsely coded using the dictionary learned by LcLcDDL method. In the feature pooling stage, the spatial pyramid matching method with max pooling is utilized to aggregate the coding coefficients of the low-level local features to generate the high-level global feature for each patch. The generated high-level global feature is feed into the linear support vector machine to recognize which classes the patch belongs to. Experimental results conducted on the moving and stationary target acquisition and recognition data set show that the performance of the proposed method is better than several available approaches.

Index Terms—Synthetic aperture radar (SAR), target recognition, attribute scattering center model (ASCM), discriminative dictionary learning, locality constrained linear coding (LLC), label consistent K-singular value decomposition (LC-KSVD), spatial pyramid matching (SPM).

I. INTRODUCTION

AS AN active sensor, synthetic aperture radar (SAR) could provide broad area image at high resolution and acquire image in inclement weather or during night as well as day. With the development of synthetic aperture technique, manual interpretation of numerous huge SAR

images is time-consuming and almost impractical [1]. A typical SAR ATR system proposed by Lincoln Laboratory consists of three stages, i.e. detection, discrimination, and classification [2], [3]. The purpose of the first two stages is to select out the chips containing the candidate targets and reject most of the clutter false alarms (CFAs). The third stage rejects the remaining CFAs and decides the target type of the selected chips, and this stage is studied in this paper.

A great many of SAR ATR algorithms have been proposed, and these algorithms can be roughly divided into template based methods [2], and model based methods [4]. Template based methods need to generate a large amount of templates, taken from the data set containing the images of targets in different poses, and the test targets are classified by matching their images with the templates. If enough training samples with all target aspect angles and deployment conditions can be obtained, it would be a great choice for target recognition. However, it is difficult to get sufficient training samples for SAR targets. Even with enough training samples, the template based methods have heavy computational burden and need huge storage space, thus, it is difficult to fast implemented these methods [5]. Therefore, researchers pay more attention on model based methods due to the robustness and effectiveness. Model based methods extract the characteristics of a data set from the physical or conceptual models of targets to predict the property of targets at different poses and deployment conditions. One kind of model based methods is called feature-based method [6], this physical model is considered to be features extracted from the original data that greatly reduces the data dimension. However, the performance of this kind model based methods heavily relies on the accuracy of the target's physical models. Another kind of model based algorithms aims to extract the target's effective features, such as the mathematical features [7], [8], and the geometrical features [9], [10]. In a word, the key point of this kind model based algorithms is the effectiveness of feature extraction. In addition, SAR ATR via neural networks (NN) [11], deep learning [12], and convolutional neural network (CNN) [13] have also been popular. However, compared with optical images, SAR images are more complex and have a smaller amount of data, it is easy to encounter overfitting and local minima if the depth network on optical image is directly applied into SAR image [14], [15]. If redesign a new depth network suitable for SAR image, there are questions, such as

Manuscript received January 4, 2019; accepted February 18, 2019. Date of publication February 22, 2019; date of current version May 16, 2019. This work was supported in part by the National Science Foundation of China under Grants U1833203, 61771362, and 61671354, and in part by the 111 Project under Grant B18039. The associate editor coordinating the review of this paper and approving it for publication was Prof. Piotr J. Samczynski. (Corresponding author: Lan Du.)

The authors are with the National Lab of Radar Signal Processing, Xidian University, Xi'an 710071, China (e-mail: dulan@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/JSEN.2019.2901050

1558-1748 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

estimating and initializing the parameters (i.e. learning rate, number of layers, convolution kernels number, etc.) for this depth network, and determining the loss function, etc.

Although the above algorithms have achieved good accuracy in SAR image recognition, only limited low-level features of the dependencies between the pixels in local region are exploited. Meanwhile, SAR image is contaminated by the strong speckle noise, and these low-level features are affected by the speckle noise. What is more, most of these low-level features are hand-crafted without consideration of the coherent imaging mechanism of SAR images.

To cope with the problems existing in these low-level features, the low-level local feature is generated by filtering the SAR patch with the attribute scattering center model (ASCM) in this paper. In this physical relevant model, an individual scattering center is described by a set of parameters characterizing its amplitude, location, shape and orientation (pose), and these parameters are strongly dependent on the geometry and material components of the target. Therefore, the ASCM is the effective key to recover physical information on the target from scattering waves incited at high frequency. According to the ASCM, several individual scattering centers can mainly make up the total back scattering response of a target. As the low-level feature is generated by convolving the ASCM (convolution kernel) with the SAR patch, when the parameters of ASCM made up the convolution kernel are the same or similar to those of a scattering center contained in the target (back scattering response), this scattering center will sharply resonate with the convolution kernel, that is, the convolution result will have a higher amplitude. Thus, there is a direct correspondence between the physical mechanisms of SAR image and the semantics of the features at low-level by the ASCM.

To thoroughly explore the discrimination power of the low-level feature, it is a way to learn or select the appropriate mid- or high-level features for SAR images. Sparse representation (SR) over a redundant dictionary is becoming a popular way of representing the data and has been proven to be efficient in the communities of signal processing and machine learning. It is assumed in SR that a sample can always be approximately by a few atoms, so that the data can be explicitly mapped into a higher order sparse feature space. Thus, SR can capture the concise structure from data, which usually contains mid- or high-level information [16], [17]. The dictionary in sparse model is quite important, and it is constituted by a redundant bases of a linear space, which can provide a discriminative feature for each sample. Benefiting from this, the sparse representation classifier (SRC) [17], [18] gives promising results in SAR target classification [19], [20] and polarimetric SAR image classification [21], [22].

Nevertheless, the dictionary in SRC is usually constructed by staking training samples, and the size of the dictionary is very big, which increases the coding complexity and the storage space. What is more, this kind dictionary may neither be optimal for solving reconstruction problems (i.e., denoising, inpainting) nor for solving discriminative problems (i.e., classification) [23]. Thus, lots of dictionary learning methods are proposed to learn a more representative and compact

dictionary. For example, Aharon *et al.* [24] propose K-SVD algorithm and Mairal *et al.* [25] put forward online dictionary learning. However, the atoms of the learned dictionary may be so highly correlated that the sparse coefficients are not stable, i.e., similar samples are represented by totally different atoms, which damages the classification performance. Discriminative KSVD (DKSVD) [26] is proposed based on the traditional K-SVD algorithm, to improve the classification ability of sparse coefficients, classification errors are incorporated into the objective functions. Label consistent K-SVD (LC-KSVD) [27] is an extension of DKSVD, which introduces a label consistent regularization to enhance the discrimination of the sparse coefficients. The LC-KSVD algorithm achieves good classification results.

As explained in [28], the nonzero coefficients of SR are usually assigned to bases near to the encoded data through empirical observation, and it also theoretically pointed out that the locality is more important under certain assumptions. Considering this and the discrimination power of the sparse coefficients, we suggest imposing both the label consistent and locality constraint on the sparse coefficients during dictionary learning process, which is called label consistent and locality constraint discriminative dictionary learning (LcLcDDL) method.

To summarize, a new SAR recognition method is proposed in this paper. It contains three main stages. In the low-level local feature extraction stage, the ASCM is used with selected parameter combinations to convolve with SAR patches. In the feature coding stage, the LcLcDDL method is proposed to generate the codebook and calculate the sparse coefficients of low-level local feature. In the feature pooling stage, the high-level global feature is generated by the spatial pyramid matching (SPM) method with max pooling. The low-level local feature is extracted with the consideration of the coherent imaging mechanism of SAR images, and the high-level global feature is then generated with label consistent knowledge and locality constraint information. Therefore, the high-level global feature is more discriminative for classification, experimental results based on MSTAR data also demonstrate this.

The remainder of this paper is organized as follows. Section II provides an overview of the proposed new method. The ASCM is described in Section III-A, and the GA and the parameter selection of the ASCM is presented in Section III-B. The instruction of the label consistent K-SVD and locality-constrained linear coding are given in Section IV-A and Section IV-B respectively, the proposed LcLcDDL method is shown in Section IV-C. The experimental results based on MSTAR are given in Section V, and Section VI concludes this paper.

II. OVERVIEW OF THE PROPOSED SAR RECOGNITION METHOD

The proposed SAR recognition method is based on the ASCM and LcLcDDL method, the flow chart of this recognition method is shown in Fig.1. All steps of this method will be described as follows.

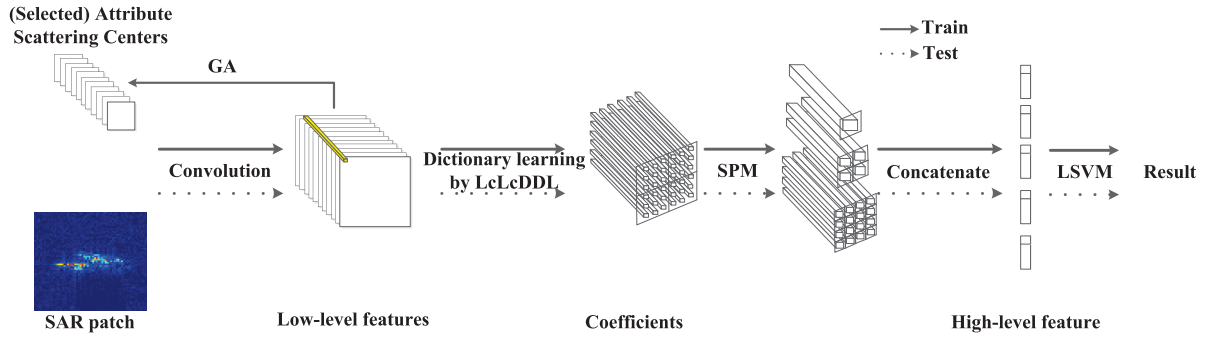


Fig. 1. The flow chart of the proposed recognition method.

1) *Low Level-Local Feature Extraction*: The low-level local features are extracted by convolving the ASCM (convolution kernel) with the SAR patches. As ASCM provides a physically relevant description of the targets, and this model includes both frequency and aspect dependence of the scattering centers derived from both physical optics and the geometric theory of diffraction. Thus, convolving with the coherent imaging SAR can extract the attribute features of SAR. What is more, the convolution process can also suppress the strong speckle noise of SAR patches.

2) *Selecting the Parameters Combinations*: The ASCM is described by seven parameters, and the possible combinations of these parameters are vast. Thus, in order to improve the discriminative power of the low-level local features, and save the training time and storage space, the genetic algorithm (GA) is applied to select the most discriminative combinations in train.

3) *Discriminative Dictionary Learning*: In order to increase the discriminative power of the sparse coefficients, both the label information and local geometric information are incorporated into the proposed dictionary learning method, LcLcDDL. The discriminative dictionary is then learned by this method with labeled patches in train.

4) *Feature Coding and Pooling*: With the learned discriminative dictionary, the sparse coefficients of the low-level local features are calculated by sparse representation. Then, the high-level global feature is generated by the spatial pyramid matching (SPM) method with max pooling.

5) *LSVM Classification*: The classifier employed in this paper is LSVM, the LSVM classifier is trained with the high-level global feature of all the training chips. The trained LSVM classifier is then used in the test stage to predict the class label of test chips.

III. THE LOW-LEVEL LOCAL FEATURE EXTRACTION BASED ON ASCM

In this section, the detailed process of the low-level local feature extraction is given. The ASCM is firstly described in the following. Then, the selection of the parameters combinations based on GA is introduced in Section III-B.

A. The Attribute Scattering Center Model (ASCM)

At high frequency, according to physical optics and geometric theory of diffraction, the total back scattering response

of an object can be approximated by a sum of responses from individual scattering centers, or scatterers [29], [30]. The back scattering field of an individual scattering center is described as a function of frequency f and aspect angle ϕ , and the total back scattering field from a target can be written as the sum of these individual scattering centers.

$$E(f, \phi; \Theta) = \sum_{i=1}^{N_0} E_i(f, \phi; \theta_i) \quad (1)$$

$$E_i(f, \phi; \theta_i) = A_i \times \left(j \frac{f}{f_c} \right)^{\alpha_i} \times \exp \left[-j \frac{4\pi f}{c} (x_i \cos \phi + y_i \sin \phi) \right] \times \text{sinc} \left(\frac{2\pi f}{c} L_i \sin(\phi - \bar{\phi}_i) \right) \times \exp(-2\pi f \gamma_i \sin \phi) \quad (2)$$

where f_c is the radar center frequency, c is the propagation velocity, N_0 is the total number of individual scattering centers, $\Theta = [\theta_1, \dots, \theta_i, \dots, \theta_{N_0}]$ represents the set of all parameters matrix of these N_0 scatterers. $E_i(f, \phi; \theta_i)$ is the response from the i th scatterer described by the parameter vector $\theta_i = [A_i, x_i, y_i, L_i, \bar{\phi}_i, \alpha_i, \gamma_i]^T$, A_i is the relative amplitude, $\alpha_i \in \{-1, -0.5, 0, 0.5, 1\}$ is the frequency and geometry dependence, the parameters x_i and y_i are the scatterer's location. According to the length L_i , scattering center can be categorized into two kinds: localized scattering center with $L_i = 0$ and distributed scattering center with $L_i \neq 0$. $\bar{\phi}_i$ is the orientation angle of distributed scattering center, γ_i is the aspect dependence of localized scattering center, $()^T$ denotes the transpose operator. Examples of scattering geometry distinguishable by their parameters (α, L) are presented in Table I [31].

Typical examples of localized scattering mechanisms are trihedral reflection, corner diffraction, and edge diffraction, and all of these scattering mechanisms have slowly varying amplitude as a function of aspect angle ϕ [29]. That is to say the aspect dependence term $\exp(-2\pi f \gamma_i \sin \phi)$ of localized scattering center in Eq.2 should be slowly varying, it means the aspect dependence γ should be small. Thus, the aspect dependence term $\exp(-2\pi f \gamma_i \sin \phi)$ of localized scattering center in Eq.2 can be regarded as a constant, and the original

TABLE I
SCATTERING GEOMETRY INTERPRETATION OF α AND L

Scattering geometry	α	L
Trihedral	1	$L = 0$
Top Hat	0.5	$L = 0$
Corner diffraction	-1	$L = 0$
Sphere	0	$L = 0$
Edge broadside	0	$L > 0$
Edge diffraction	-0.5	$L > 0$
Dihedral	1	$L > 0$
Cylinder	0.5	$L > 0$

ASCM described in Eq.2 can be simplified as follows:

$$E_i(f, \phi; \theta_i) = A_i \times \left(j \frac{f}{f_c} \right)^{\alpha_i} \times \exp \left[-j \frac{4\pi f}{c} (x_i \cos \phi + y_i \sin \phi) \right] \times \text{sinc} \left(\frac{2\pi f}{c} L_i \sin(\phi - \bar{\phi}_i) \right) \quad (3)$$

B. Parameter Selection of ASCM Based on GA

Although Eq.2 can be simplified as Eq.3 by regarding the aspect dependence term as a constant, and the relative amplitude A_i can be set as $A_i = 1$, there are still five other parameters need to be defined. It is too difficult to deal with the thousands possible combinations of these five parameters. Thus, in order to select the most discriminative combinations of ASCM parameters, the genetic algorithm is introduced.

GA is developed by John Holland at the University of Michigan in the early 1970s [32]. GA is one of the heuristic-based optimization techniques and successfully applied in many optimization problems. In GA, a population of strings (called chromosomes or the genotype of the genome), which encode candidate solutions (called individuals, creatures, or phenotypes) to an optimization problem, evolves toward better solutions. The evolution starts from generating a group of random solutions (individuals), each individuals contains several properties (chromosomes), then, based on the laws of genetics, selection, crossover and mutation occur in chromosomes to form a second generation of individuals with more diverse properties. The new generation is then used in the next iteration of GA. Commonly, GA terminates when either a satisfactory fitness level has been reached for the population or a maximum number of generations have been produced.

1) *Population Initialization*: In order to define an optimal way to initialize the population, the latin hypercube sampling (LHS) method is used [33]. The method is used to generate semi-random samples in the parameter domain, in such a way, the parameter domain is optimally sampled to generate a population whose values are extended in a given domain and these individuals fill the domain in the optimal way.

2) *Selection Operator*: In order to create new individuals, the individuals with better fitness in the current generation must be selected to transfer their genes to the next generations. This process allows to select the suitable genes and to eliminate the worst ones. There are many different methods in the selection process, but in order to add randomness to GA, a stochastic universal sampling (SUS) method [34] is used. This method selects individuals from the group according to their fitness, so that all individuals can be selected, but it is more likely to choose the best-fitted individuals.

3) *Crossover Operator*: After selection, in order to produce new and better individuals, the previously selected individuals will be recombined to generate offspring with better properties in principle. Then, a crossover point along each chromosome is chosen followed by the exchange of the values up to the crossover point between the two chromosomes, and this process of crossover will be iterated over and over until the desired diversity of individuals is made.

4) *Mutation Operator*: It is an important operator for the reason that thanks to the mutation, some genes can take values that are impossible if without the mutation. With this genetic operator, the parametric domain can be scanned in a better way. In GA, mutation is the last operator and any gene of each individual in the new generation could be mutated.

5) *Fitness Evaluation*: The algorithm must evaluate the fitness level of the new generation to select the following candidates whose genes will be recombined. In this paper, the fitness can be explained as the discriminative power, because the GA is used as a selection method to select the most discriminative combinations of ASCM parameters. The ratio of the between-class distance to the within-class distance (RBTW) [35] is used to measure the fitness level (discriminative power) of the new generations, an ideal individual will achieve the bigger RBTW value.

$$\dot{\mathbf{x}}_k^l = nz(\mathbf{w} \odot \mathbf{x}_k^l) \quad (4)$$

$$S_B = \frac{1}{N} \sum_{l=1}^L n_l \times (\bar{\mathbf{x}}^l - \bar{\mathbf{x}})^T (\bar{\mathbf{x}}^l - \bar{\mathbf{x}}) \quad (5)$$

$$S_W = \frac{1}{N} \sum_{l=1}^L \sum_{k=1}^{n_l} n_l \times (\dot{\mathbf{x}}_k^l - \bar{\mathbf{x}}^l)^T (\dot{\mathbf{x}}_k^l - \bar{\mathbf{x}}^l) \quad (6)$$

$$J = S_B / S_W \quad (7)$$

where $\mathbf{x}_k^l$ denotes the k th data belonging to l th class, and each entry of $\mathbf{x}_k^l$ corresponding to an combination of ASCM parameters. $\mathbf{w}$ is an individual of GA indicating the selected combinations, the values in $\mathbf{w}$ are either 1 or 0, $\odot$ denotes the element-wise multiplication, $nz()$ means to recombination the vector with the non-zero entries. $\bar{\mathbf{x}}^l$ is the mean value of all samples from l th class and $\bar{\mathbf{x}}$ represents the mean value of samples from all classes. L denotes the number of classes, n_l denotes the number of samples from l th class, and N is the total number of all samples, J is the RBTW value. S_B and S_W are the between-class distance and within-class distance respectively. The larger between-class distance means that classes are far away from each other. The smaller within-class

distance means the samples are close to each other in the category.

IV. LABEL CONSISTENT AND LOCALITY CONSTRAINT DISCRIMINATIVE DICTIONARY LEARNING (LcLcDDL)

As explained in Section I, with the aim to incorporate both the label information and local geometric information into the sparse representation, the LcLcDDL method is proposed. In this section, the LC-KSVD and locality constrained linear coding (LLC) [36] are firstly brief introduced, then, we describe the proposed LcLcDDL method.

A. Label Consistent K-SVD (LC-KSVD)

Jiang *et al.* [27] propose to combine a new label consistent constraint, ‘discriminative’ sparse code error with the reconstruction error or classification error, encouraging the similarity among the sparse coefficients of samples belonging to the same class. The authors claimed that this additional term improves the accuracy of the linear classifier obtained by DKSVD [26]. The LC-KSVD is formulated as:

$$\begin{aligned} (\mathbf{B}^*, \mathbf{A}^*, \mathbf{C}^*) = \arg \min_{\mathbf{B}, \mathbf{A}, \mathbf{C}} & \|\mathbf{X} - \mathbf{B}\mathbf{C}\|_2^2 + \beta \|\mathbf{Q} - \mathbf{A}\mathbf{C}\|_2^2 \\ \text{s.t. } \forall i, & \|\mathbf{c}_i\|_0 < T_0 \end{aligned} \quad (8)$$

where $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_K] \in R^{M \times K}$ is a set of M -dimensional labeled training samples, and the number of this training samples is K , $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_N] \in R^{M \times N}$ is the learned redundant dictionary with $N > M$, $\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_K] \in R^{N \times K}$ are the sparse coefficients of input training samples $\mathbf{X}$, $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_K] \in R^{N \times K}$ are the ‘discriminative’ sparse codes corresponding to the original training samples $\mathbf{X}$ for classification, and $\mathbf{q}_k$ is the ‘discriminative’ sparse code corresponding to input sample $\mathbf{x}_k$, $\mathbf{A} \in R^{N \times N}$ is a linear transformation matrix that converts the sparse coefficients in $\mathbf{C}$ to be more discriminative in sparse feature space, T_0 is sparsity constraint factor, β is the regularization parameter to balance the ‘discriminative’ sparse code errors and the reconstruction error.

In Eq.8, $\mathbf{Q}$ is defined as an ideal ‘discriminative’ sparse code matrix if the nonzero values of $\mathbf{q}_k$ occur at those indices where input sample $\mathbf{x}_k$ and the dictionary atom $\mathbf{b}_n$ share the same label. Specifically, supposing $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5, \mathbf{b}_6]$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6, \mathbf{x}_7, \mathbf{x}_8]$, where $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{b}_1$ and $\mathbf{b}_2$ are from class 1, $\mathbf{x}_4, \mathbf{x}_5, \mathbf{b}_3$ and $\mathbf{b}_4$ are from class 2, and $\mathbf{x}_6, \mathbf{x}_7, \mathbf{x}_8, \mathbf{b}_5$ and $\mathbf{b}_6$ are from class 3, $\mathbf{Q}$ is defined as:

$$\mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \quad (9)$$

B. Locality-Constrained Linear Coding (LLC)

The locality-constrained linear coding algorithm [36] has been developed as an effective coding method for classification. Instead of sparsity, the LLC tends to use the dictionary

atoms that are near (similar) to the input samples to represent the input itself. In other words, nearby or local atoms of the input sample are used to code the input sample. The LLC objective function is:

$$\begin{aligned} (\mathbf{B}^*, \mathbf{C}^*) = \arg \min_{\mathbf{B}, \mathbf{C}} & \sum_{i=1}^K \|\mathbf{x}_i - \mathbf{B}\mathbf{c}_i\|_2^2 + \lambda \|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2 \\ \text{s.t. } \forall i, & \mathbf{1}^T \mathbf{c}_i = 1 \end{aligned} \quad (10)$$

where $\mathbf{d}_i \in R^{N \times 1}$ is the locality adaptor which is proportion to the similarity between bases and the input sample $\mathbf{x}_i$, λ is a parameter controlling the locality constraint. The constraint $\mathbf{1}^T \mathbf{c}_i = 1$ follows the shift-invariant requirements of the LLC coefficients. Note that the LLC coefficient in Eq.10 is not sparse in the sense of l_0 norm, but is sparse in the sense that the solution has only a few significant values. In practice, those small coefficients are simply set to be zeros.

The term $\|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2$ is the locality constraint which projects each sample into its local coordinate system. The locality adaptor $\mathbf{d}_i$ is defined as:

$$\mathbf{d}_i = \exp\left(\frac{\text{dist}(\mathbf{x}_i, \mathbf{B})}{\sigma}\right) \quad (11)$$

where $\text{dist}(\mathbf{x}_i, \mathbf{B}) = [\|\mathbf{x}_i - \mathbf{b}_1\|_2, \dots, \|\mathbf{x}_i - \mathbf{b}_N\|_2]^T$, and σ is used to adjust the weight decay speed for the locality adaptor.

An attractive property of LLC is its sparse coefficient can be derived analytically by:

$$\tilde{\mathbf{c}}_i = (\mathbf{C}_i + \lambda \text{diag}^2(\mathbf{d}_i)) \setminus \mathbf{1} \quad (12)$$

$$\mathbf{c}_i = \tilde{\mathbf{c}}_i / (\mathbf{1}^T \tilde{\mathbf{c}}_i) \quad (13)$$

where $\mathbf{C}_i = (\mathbf{B}^T - \mathbf{1}\mathbf{x}_i^T)(\mathbf{B}^T - \mathbf{1}\mathbf{x}_i^T)^T$ denotes the data covariance matrix. What is more, an approximation can be adopted to accelerate the computational efficiency in practice. Instead of solving Eq.10, we can solve a much smaller linear system to get the coefficients:

$$\begin{aligned} \hat{\mathbf{C}}^* = \arg \min_{\hat{\mathbf{C}}} & \sum_{i=1}^K \|\mathbf{x}_i - \mathbf{B}_i \hat{\mathbf{c}}_i\|_2^2 \\ \text{s.t. } \forall i, & \mathbf{1}^T \hat{\mathbf{c}}_i = 1 \end{aligned} \quad (14)$$

where $\mathbf{B}_i$ is the local bases containing T ($T < M < N$) nearest neighbors of $\mathbf{x}_i$, $\hat{\mathbf{c}}_i$ and $\hat{\mathbf{C}}^*$ are the approximate solution of $\mathbf{c}_i$ and $\mathbf{C}^*$ respectively.

C. Label Consistent and Locality Constraint Discriminative Dictionary Learning (LcLcDDL)

Both the label information and local geometric information are helpful for classification, thus the ‘discriminative’ sparse code error, the locality constraint and the reconstruction error are combined together to learn a more discriminative dictionary. Thus, the LcLcDDL method is formulated as:

$$\begin{aligned} (\mathbf{B}^*, \mathbf{A}^*, \mathbf{C}^*) = \arg \min_{\mathbf{B}, \mathbf{A}, \mathbf{C}} & \sum_{i=1}^K \|\mathbf{x}_i - \mathbf{B}\mathbf{c}_i\|_2^2 + \beta \|\mathbf{q}_i - \mathbf{A}\mathbf{c}_i\|_2^2 \\ & + \lambda \|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2 \quad \text{s.t. } \forall i, \mathbf{1}^T \mathbf{c}_i = 1 \end{aligned} \quad (15)$$

where the term $\|\mathbf{x}_i - \mathbf{B}\mathbf{c}_i\|_2^2$ represents the reconstruction error, which restrains the linear combinations of the learned dictionary to be more similar to the original input. The term $\|\mathbf{q}_i - \mathbf{A}\mathbf{c}_i\|_2^2$ represents the ‘discriminative’ sparse code error, which enforces the input from the same class to have similar sparse coefficients. The last term $\|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2$ is the locality constraint which restricts each input into its local coordinate system.

Inspired by the optimization method described in [27], Eq.15 can be rewritten as:

$$\langle \mathbf{B}^*, \mathbf{A}^*, \mathbf{C}^* \rangle = \arg \min_{\mathbf{B}, \mathbf{A}, \mathbf{C}} \sum_{i=1}^K \left\| \begin{pmatrix} \mathbf{x}_i \\ \sqrt{\beta} \mathbf{q}_i \end{pmatrix} - \begin{pmatrix} \mathbf{B} \\ \sqrt{\beta} \mathbf{A} \end{pmatrix} \mathbf{c}_i \right\|_2^2 + \lambda \|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2 \quad s.t. \quad \forall i, \mathbf{1}^T \mathbf{c}_i = 1 \quad (16)$$

Let $\mathbf{y}_i = (\mathbf{x}_i, \sqrt{\beta} \mathbf{q}_i)^T$, $\mathbf{y}_i \in R^{(M+N) \times 1}$, $\mathbf{D} = (\mathbf{B}, \sqrt{\beta} \mathbf{A})^T$, $\mathbf{D} \in R^{(M+N) \times N}$, the matrix $\mathbf{D}$ is then l_2 normalized column wise. The optimization of Eq.16 is equivalent to solve the following problems:

$$\langle \mathbf{D}^*, \mathbf{C}^* \rangle = \arg \min_{\mathbf{D}, \mathbf{C}} \sum_{i=1}^K \|\mathbf{y}_i - \mathbf{D}\mathbf{c}_i\|_2^2 + \lambda \|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2 \quad s.t. \quad \forall i, \mathbf{1}^T \mathbf{c}_i = 1 \quad (17)$$

The sparse coefficient of Eq.17 can also be solved analytically like the LLC method, the only difference is the data covariance matrix. In here, this matrix is denoted as $\mathbf{C}_i = (\mathbf{D}^T - \mathbf{1}\mathbf{y}_i^T)(\mathbf{D}^T - \mathbf{1}\mathbf{y}_i^T)^T$. The method to obtain the approximation solution of Eq.17 is almost the same as LLC. Finding T nearest neighbors of $\mathbf{y}_i$ as the local bases $\mathbf{D}_i$, and solving the following linear system:

$$\hat{\mathbf{C}}^* = \arg \min_{\hat{\mathbf{C}}} \sum_{i=1}^K \|\mathbf{y}_i - \mathbf{D}_i \hat{\mathbf{c}}_i\|_2^2 \quad s.t. \quad \forall i, \mathbf{1}^T \hat{\mathbf{c}}_i = 1 \quad (18)$$

Note that, experimental results shows that the analytical solution and the approximation solution of Eq.17 have little effect on the classification accuracy, but the approximation solution is calculated much faster than the analytical one. Thus, only the approximation solution of Eq.17 is used in experiments.

As for the dictionary $\mathbf{D}$, the incremental dictionary learning method is utilized. The process of this method is elaborately described here, the dictionary $\mathbf{B}$ and the linear transformation matrix $\mathbf{A}$ are firstly initialized, and the initialization method will be described in following. Then the dictionary $\mathbf{D} = (\mathbf{B}, \sqrt{\beta} \mathbf{A})^T$ is updated in an incremental manner with all the training samples. In each iteration, a single training sample $\mathbf{x}_i$ and its corresponding ‘discriminative’ sparse codes $\mathbf{q}_i$ are used to contact as $\mathbf{y}_i = (\mathbf{x}_i, \sqrt{\beta} \mathbf{q}_i)^T$. Solve Eq.17 to obtain the corresponding sparse coefficient $\mathbf{c}_i$, and a new sub dictionary $\mathbf{D}_{sub}$ is formed by selecting all the atoms whose corresponding weights are larger than a predefined threshold. Then, refit sample $\mathbf{y}_i$ without the locality constraint as Eq.18. The obtained code is then used to update the sub dictionary $\mathbf{D}_{sub}$ in a gradient descent fashion. Finally, we project those atoms outside the unit circle onto the unit circle. The above process is illustrated in Algorithm 1.

Algorithm 1 Incremental Dictionary Learning

Input: The training samples $\mathbf{x}_i$, $i = 1, \dots, K$, the corresponding ‘discriminative’ sparse code $\mathbf{q}_i$, the initialization of dictionary $\mathbf{B}$, the linear transformation matrix $\mathbf{A}$, the parameters σ , β , λ

```

1 for  $i = 1$  to  $K$  do
2   Let  $\mathbf{y}_i = (\mathbf{x}_i, \sqrt{\beta} \mathbf{q}_i)^T$ ,  $\mathbf{D} = (\mathbf{B}, \sqrt{\beta} \mathbf{A})^T$ , and initialize  $\mathbf{d}_i = [d_1, \dots, d_j, \dots, d_N]$ ;
3   for  $j = 1$  to  $N$  do
4     Computing the similarity between sample  $\mathbf{x}_i$  and each dictionary atom  $\mathbf{b}_j$ :  $d_j = \exp\left(\frac{\text{dist}(\mathbf{x}_i, \mathbf{b}_j)}{\sigma}\right)$ ;
5   end
6   Coding  $\mathbf{c}_i = \arg \min_{\mathbf{c}_i} \|\mathbf{y}_i - \mathbf{D}\mathbf{c}_i\|_2^2 + \lambda \|\mathbf{d}_i \odot \mathbf{c}_i\|_2^2$ ,  $s.t. \mathbf{1}^T \mathbf{c}_i = 1$  using Eq.12 and Eq.13 with  $\mathbf{C}_i = (\mathbf{D}^T - \mathbf{1}\mathbf{y}_i^T)(\mathbf{D}^T - \mathbf{1}\mathbf{y}_i^T)^T$ ;
7   Selecting some atoms to form a new sub-dictionary according to:  $id = \{k | abs(\mathbf{c}_i(k)) > 0.01\}$ ,  $\mathbf{D}_{sub} = \mathbf{D}(:, id)$ ;
8   Re-coding over sub dictionary  $\mathbf{D}_{sub}$ :  $\hat{\mathbf{c}}_i = \arg \min_{\mathbf{c}_i} \|\mathbf{y}_i - \mathbf{D}_{sub}\mathbf{c}_i\|_2^2$ ,  $s.t. \mathbf{1}^T \mathbf{c}_i = 1$ ;
9   Updating  $\mathbf{D}_{sub} = \mathbf{D}_{sub} - \frac{\mu \Delta \mathbf{D}_{sub}}{\|\hat{\mathbf{c}}_i\|_2}$ , where  $\Delta \mathbf{D}_{sub} = -2\hat{\mathbf{c}}_i(\mathbf{y}_i - \mathbf{D}_{sub}\hat{\mathbf{c}}_i)$  and  $\mu = 1/\sqrt{i}$ ;
10  Project the sub dictionary  $\mathbf{D}_{sub}$  to dictionary  $\mathbf{D}$ :  $\mathbf{D}(:, id) = \text{proj}(\mathbf{D}_{sub})$ ;
11 end
Output: The learned dictionary  $\mathbf{B} = \text{proj}(\mathbf{D}(1:M, :))$ 

```

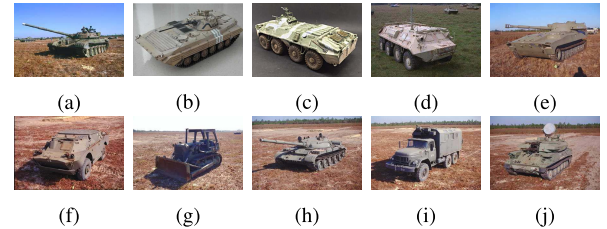


Fig. 2. Optical images of the ten-target data. (a) T72. (b) BMP2. (c) BTR70. (d) BTR60. (e) 2S1. (f) BRDM2. (g) D7. (h) T62. (i) ZIL131. (j) ZSU234.

Here, we discuss the initialization of the dictionary $\mathbf{B}$ and the linear transformation matrix $\mathbf{A}$. As for $\mathbf{B}$, several iterations of K-SVD [24] are employed within each class and then all the outputs of each K-SVD are combined together. The label of each dictionary atom is initialized based on the class it corresponds to and will remain fixed during the entire dictionary learning process, although the atom is updated during the learning process. Note that, although each dictionary atom is associated a unique and fixed class label, an input sample of a class certainly can use dictionary atoms from other classes.

To initialize matrix $\mathbf{A}$, the multivariate ridge regression model [37] is employed, as follows:

$$\mathbf{A} = \arg \min_{\mathbf{A}} \|\mathbf{Q} - \mathbf{A}\mathbf{C}\|_2^2 + \lambda_1 \|\mathbf{A}\|_2^2 \quad (19)$$

the solution is:

$$\mathbf{A} = \mathbf{Q}\mathbf{C}^T (\mathbf{C}\mathbf{C}^T + \lambda_1 \mathbf{I})^{-1} \quad (20)$$

Given the initialized $\mathbf{B}$, the original K-SVD is applied to compute the sparse codes $\mathbf{C}$ of the training samples $\mathbf{X}$. Then, $\mathbf{C}$ can be used to compute the initial $\mathbf{A}$ in Eq.20. λ_1 is just a balance parameter.

When the dictionary $\mathbf{B}$ is well learned by Algorithm 1, for a test sample $\mathbf{z}$, its spares coefficient is computed through Eq.10 with learned dictionary $\mathbf{B}$.

V. EXPERIMENTAL RESULTS

To verify the performance of the proposed recognition method, the following experiments are designed. The effectiveness of the low-level local feature and the LcLcDDL method are firstly tested. Subsequently, in order to determine the parameters of the proposed method and achieve the best target recognition performance, the influences of the parameters are researched. Finally, the recognition results and analysis are given.

A. Introduction of MSTAR Data Set

The experiments are conducted on the MSTAR public release data set, this data set is collected by the Sandia National Laboratory using a Twin Otter SAR sensor operating at X-band HH polarization with $0.3m \times 0.3m$ resolution (As the patches used in this paper is without windowing, the actual resolution is $0.2m \times 0.2m$), comprising a various of military terrain target images and background cluster taken at multiple depression angles (15° , 17° , 30° , 45°) over a full $0^\circ - 360^\circ$ range of viewing angles. As far as we know, this is the only publicly published data set that can be used for SAR target recognition. Following previous research, 10 kinds of targets will be utilized in this paper, and the optical images of these 10 targets are displayed in Fig 2.

As explained in Section II, the SPM method is used in the feature pooling stage to generate the high-level global feature. Here, we will give more detailed description of SPM. Motivated by Grauman and Darrell [38], the SPM method partitions the image into increasingly finer spatial sub-regions and computes histograms of local features from each sub-region. Typically, $2^l \times 2^l$ sub-regions, $l = 0, 1, 2$ are used, we also use these typical parameters in this paper. Following the work in [18] and [36], the max pooling is used to aggregate the low-level local feature coefficients in each sub-region, the aggregated feature in each sub-region is concatenated and normalized with the l_2 norm to form the high-level global feature of the SAR chip.

B. Effectiveness of the Low-Level Local Feature

As explained in Section II and shown in Fig.1, the ASCM is utilized to cope with the strong speckle noise of SAR patches and get attribute feature of SAR patches. The low-level local feature is generated by convoluting through the SAR patches with convolution kernel made up by ASCM. Firstly, a convolution kernel is produced by ASCM as Eq.3 with one of the selected parameter combinations (SPCs), the SPCs will be

following described. Then, convolve this special convolution kernel through one SAR patch generating the convoluted patch. Repeat these two processes until all the SPCs are used. The values of the same location in these convoluted patches are connected to form a low-level local feature, thus one SAR patch will produce lots of low-level local features.

Here, we will firstly describe the original parameters combinations (OPCs) of ASCM. In Section III-A, the Eq.2 is simplified as Eq.3 by regarding the aspect dependence term $\exp(-2\pi f \gamma_i \sin \phi)$ of localized scattering center as a constant, i.e. setting $\gamma_i = 0$ in Eq.2. The influence of different value of A_i is just the amplitude of the convolution response, and the amplitude is in direct proportion to A_i , there is no additional classification information is introduced or reduced (the same to low-level feature and high-level feature). Thus, the relative amplitude can be set as $A_i = 1$. The frequency dependence α_i is predefined by the ASCM as $\{-1, -0.5, 0, 0.5, 1\}$. Considering the resolution of MSTAR data set and the size of the military terrain targets used in this paper, the size of the convolution kernel is set as 30×30 ($6m \times 6m$). Because the convolution kernel will convolve through the SAR patches, it does not matter where the scattering center locates in the convolution kernel. Thus, the attribute scattering center is set to locate in the center of the convolution kernel, i.e. $x_i = y_i = 15$, x_i and y_i are the down range and cross range location of this scattering center. Considering the practical situation and the parameter of MSTAR data set, the length L_i of the distributed scattering center is set as $[0, 30\delta_A]$ with δ_A interval, δ_A is the azimuth resolution ($\delta_A = 0.2m$), as for the orientation angle $\bar{\phi}_i$ of distributed scattering center is set as $[-3^\circ, 3^\circ]$ with 0.5° interval. When the parameters scope of ASCM is given, then combining all these available parameters forms the OPCs of ASCM.

During the training stage, the OPCs will be selected by the GA algorithm. The selected parameter combinations are aforementioned SPCs. The SPCs will be directly used in the testing stage.

In addition, the size of the MSTAR data set are not the same, to avoid the influence of the clutter and following other recognition framework, the middle block with the size of 93×93 ($18.6m \times 18.6m$) for each data is used in the following experiments. Because the convolution results are computed without zero-padding in this paper, using this operation, the size of the final convoluted patch is $(93 - 30 + 1) \times (93 - 30 + 1)$, i.e. 64×64 ($12.8m \times 12.8m$), the same size as most of [10] and [39].

To evaluate the effectiveness of the low-level local feature, the low-level local features are feed in to SPM directly to generate the global feature of patches. The low-level local features are generated with OPCs and SPCs respectively. Here, the scale-invariant feature transform (SIFT) feature [40] is adopt as the baseline. The three targets recognition results are shown in Table II, the three targets used in here will be described in Section V-E.

Table II shows the recognition results in term of the probability of correct classification P_{cc} , which is defined as the percentage of targets that are correctly classified. (If not specified in the following, the recognition results are

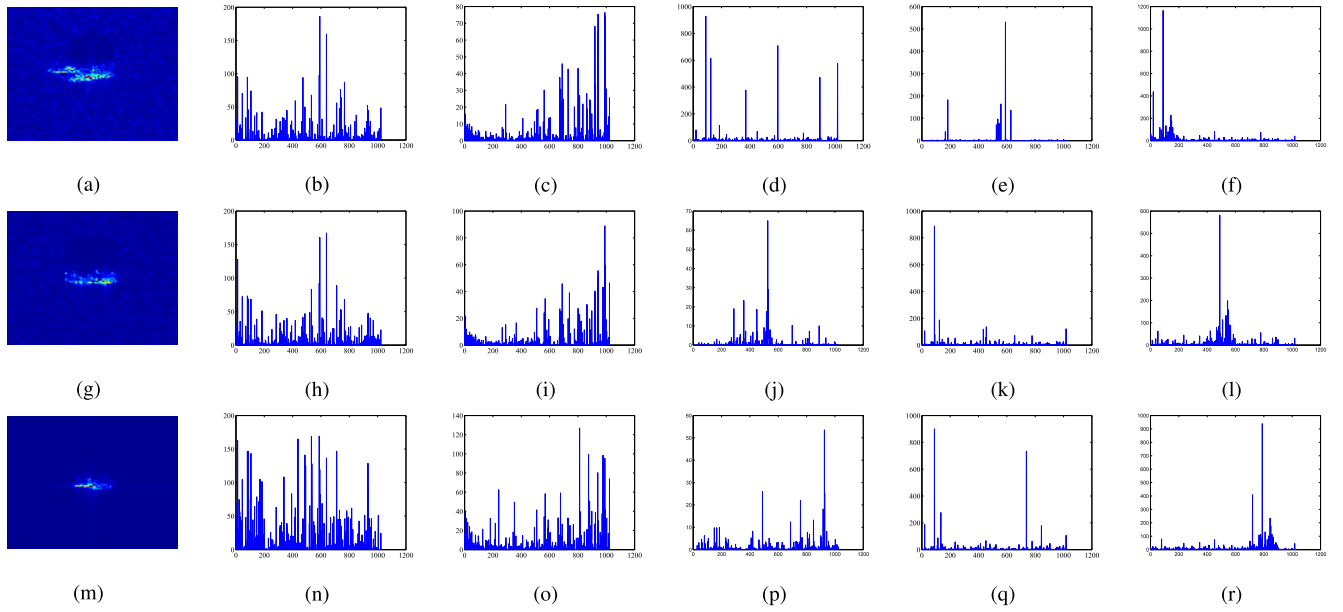


Fig. 3. Examples of sparse coefficients using different dictionary learning methods on three different MSTAR targets. Each waveform indicates a sum of absolute sparse coefficients for different test samples. The curves in the first, second and third rows corresponding to T72, BTR70, BMP2, respectively. The first column is the samples from MSTAR data set, the second, third, fourth, fifth and sixth column is the sparse coefficients using SRC, K-SVD, LC-KSVD, LLC and LcLcDDL method.

TABLE II
DETAILED ACCURACIES OF DIFFERENT FEATURES
ON THE THREE TARGETS DATASET

Features	T72	BTR70	BMP2	Avg
SIFT	88.83%	95.41%	75.98%	84.25%
OPCs	92.96%	86.22%	75.98%	84.69%
SPCs	90.05%	90.78%	87.14%	88.90%
	$\pm 0.92\%$	$\pm 1.37\%$	$\pm 1.22\%$	$\pm 0.62\%$

all represented by P_{cc}). In Table II, ‘SIFT’ represents the low-level local features generated by dense SIFT descriptor, ‘OPCs’ represents the low-level local features generated with OPCs of ASCM, ‘SPCs’ represents the low-level local features generated with SPCs of ASCM. Note that, as GA adopts random initialization, the SPCs may be different between two experiments. Thus, we do this experiment 50 times to evaluate the influence of random initialization. The mean and variance of these 50 experiments are shown in the last row of Table II. From these results, we can see that the variance is small, that is to say that the recognition results are more stable even with different SPCs. For simply, the following experiments just randomly do once. As shown in Table II, we can see that the low-level local feature generated with ASCM is superior to the SIFT feature, this result shows that the low-level local feature generated with ASCM can be used as SAR recognition feature. Moreover, the average rate of ‘SPCs’ is higher than that of ‘OPCs’, which indicates that the high dimensional low-level local features generated with OPCs of ASCM is redundant, and the GA selects the most discriminative parameter combinations of ASCM.

C. Effectiveness of the LcLcDDL Method

To learn a more discriminative dictionary, a dictionary learning method called LcLcDDL is proposed in Section IV-C. This dictionary learning method incorporates both the label information and local geometric information into the sparse representation. In here, we will discuss the effectiveness of the LcLcDDL method from two aspects. Firstly, the coefficients of different MSTAR targets will be shown in Fig.3. Secondly, the coefficients of different MSTAR targets will be projected to two-dimensional PCA subspace to compare the discrimination ability qualitatively.

In the following, SRC [17], K-SVD [24], LC-KSVD [27], and LLC [36] are adopted to compare with the proposed LcLcDDL.

From Fig.3, we can see that there is little difference in the SRC coefficients and K-SVD coefficients of these three targets. As the label information used in the LC-KSVD and LcLcDDL in training, most of the sparse coefficients of these two methods spread in the corresponding dictionary atoms, i.e. the test samples are linear combined with the atoms labeled with the same category as the input samples. Note that, when learn the sparse coefficients of the test samples, there is no label information used. What is more, as the locality constraint used in LcLcDDL, the sparse coefficients of LcLcDDL are more similar than those of LC-KSVD (the strong coefficients of LcLcDDL are more concentrated). From Fig.4, the learned coefficients by the proposed LcLcDDL method are obviously more discriminative than those by other comparison methods.

D. Algorithm Analysis

In this part, we will analyse the proposed recognition algorithm based on the ASCM and LcLcDDL method.

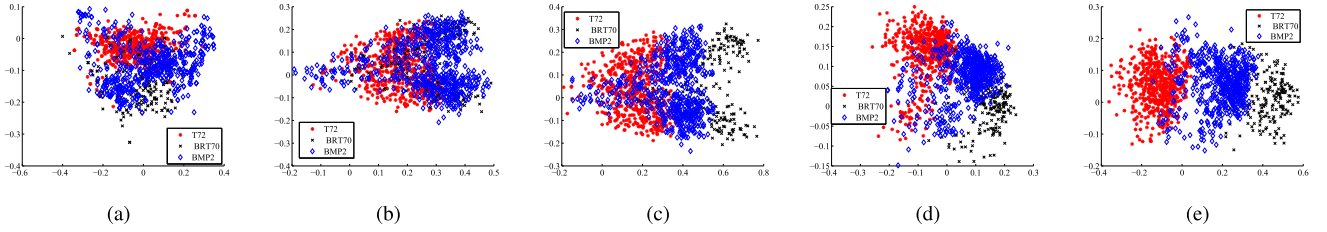


Fig. 4. Two-dimensional PCA projection of MSTAR training samples: (a) is learned by SRC, (b) is learned by K-SVD, (c) is learned by LC-KSVD, (d) is learned by LLC, and (e) is learned by LcLcDDL.

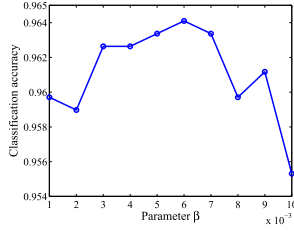


Fig. 5. Accuracies with different β .

The analysis of parameters setting, the algorithm complexity and the time consuming will be given in the following. All the experiments in this part are conducted on the three targets recognition data set, which will be described in Section V-E.

1) *Parameters Setting*: The LcLcDDL method proposed in Section IV-C includes five parameters that can be adjusted: the weight decay speed parameter σ , the balance parameters β and λ , the dictionary size N , and an implicit parameter sparsity T .

The parameters σ and λ in in Eq.(11) and Eq.15 are two related parameters, which need to be carefully selected that the length of id in line 7 of Algorithm 1 can match with the desired sparsity T . Finally, $\sigma = 100$ and $\lambda = 0.1$ are used to meet the desired sparsity $T = 15$ (This sparsity is selected according to the following experiments.). What is more, the parameters β and λ are to adjust the reconstruction error, the ‘discriminative’ sparse code error and the locality constraint to the same order of magnitude. The calculated average values of these three parts are 0.052, 8.394 and 0.304 respectively. Here, the initialization of sparse code is same to Eq.20. As discussed before, $\lambda = 0.1$ together with $\sigma = 100$ is selected to meet the desired sparsity. Fig.5 shows the variation of the classification accuracy of our method with the value of parameter $\beta \in [0.001, 0.01]$. We can see that the optimal value is $\beta = 0.06$, and either larger or smaller value of β decreases the classification accuracy. A too large value of β means a tight restriction to force the input from the same class to have similar sparse coefficients in train, it is not good for generalization. Otherwise, a too small one is not able to provide enough restriction. Same experiments can be conducted on ten target data set, and the optimal values of these parameters are $\sigma = 100$, $\beta = 0.05$ and $\lambda = 0.1$.

As for the other two parameters: the dictionary size N , and the sparsity T , we conduct several experiments to evaluate the impact of these two parameters on the classification accuracy

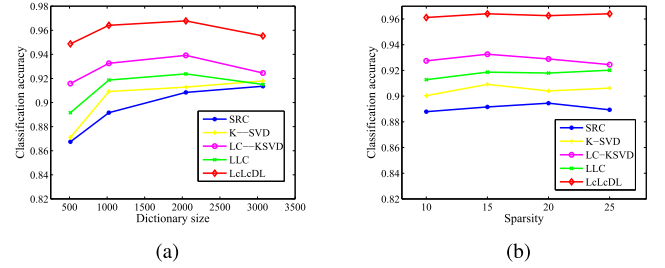


Fig. 6. The classification accuracy versus the dictionary size and the sparsity. (a) The classification accuracy vs. dictionary size. (b) The classification accuracy vs. sparsity.

performance. Different dictionary sizes $N = 512, 1024, 2048$, and 3072 with $T = 15$, and different sparsity $T = 10, 15, 20$, and 25 with $N = 1024$ are evaluated. The experiments results are shown in Fig.6.

As shown in Fig.6(a), our approach maintains high classification accuracy and outperforms the other competing approaches significantly, even using a smaller dictionary size. The basic reason for the good recognition performance is that the learned dictionary encourages the samples from the same class to have similar and congregated sparse coefficients, and those from different classes to have dissimilar sparse coefficients, which is very important for linear classification. Fig.6(b) shows that the performance of all approaches change little when the sparsity increases. The possible reason is that the high sparsity may cause high correlation between sparse coefficients, and this high correlation may not introduce useful information for increasing classification accuracy. Experiments about these two parameters: the dictionary size N , and the sparsity T , on ten targets classification dataset produce the similar results, and the detail experiment results will not be given due to space limitation.

2) *Algorithm Complexity Analysis*: In this part, the complexity of the proposed recognition method will be discussed. In the test stage, the main processes are the low-level local feature extraction and learning the sparse coefficients of the low-level local features. The low-level local features are generated by convolution and the computation complexity of the convolution process is $o(n_1^2 n_2^2)$. Here, n_1 is the size of the convolution kernel, i.e. $n_1 = 30$, n_2 is the size of the final convoluted patch, i.e. $n_2 = 64$. The computation complexity of learning the approximation spares coefficients is $o(N + T^2)$ (the same as LLC method), N is the dictionary

TABLE III
AVERAGE TEST TIME OF DIFFERENT METHODS

Methods	Test time(s)
SRC	1.26
K-SVD	1.68
LC-KSVD	1.72
LLC	1.08
LcLcDDL	1.11

TABLE IV
TARGETS TYPE AND NUMBER OF TRAINING AND TEST SAMPLES
FOR THREE TARGETS RECOGNITION

Targets	T72			BRT70	BMP2		
	132	S7	812		C21	9566	9563
Train 17°	232	0	0	233	233	0	0
Test 15°	196	191	195	196	196	196	195

TABLE V
DETAILED ACCURACIES OF DIFFERENT METHODS
ON THREE TARGETS DATASET

Methods	T72	BTR70	BMP2	Avg
SRC	89.69%	90.82%	88.07%	89.16%
K-SVD	91.92%	93.88%	88.93%	90.92%
LC-KSVD	92.78%	96.43%	92.67%	93.26%
LLC	92.10%	94.39%	90.80%	91.87%
LcLcDDL	96.22%	97.96%	96.08%	96.41%

size, T is the sparsity, and $T \ll N$ in practical. Thus, the main computational complexity of the proposed recognition method is $o(n_1^2 n_2^2 + N + T^2)$.

To compare, we also give the computation complexity of LC-KSVD (same as K-SVD) and SRC in learning the sparse coefficients. The orthogonal matched pursuit (OMP) is used to obtain the sparse coefficients when dictionary is learned, thus the computation complexity is $o(M^3 T)$, M is the dimensional of the low-level features. Solving the sparse coefficients of SRC usually requires optimization procedures, OMP can also be used. Another method, the Feature Sign algorithm utilized by Yang *et al.* [18] is used in here, and the computation complexity is $o(NT)$ in the optimal case [41].

3) *Time Consuming Analysis*: The average test time of the proposed LcLcDDL method and other comparison methods: SRC, K-SVD, LC-KSVD, and LLC are listed in Table III. The experiments of all methods are conducted using a PC with Intel(R) Core i5-3570 CPU of 3.4 GHz and memory of 4 GB. The program codes are written in MATLAB. The average test time of the proposed LcLcDDL method is lower, comparable with LLC method, this makes the LcLcDDL method suitable in many real applications.

E. Recognition Results and Analysis

1) *Three Targets Recognition Results and Analysis*: In this three targets experiment, both three distinct types of targets: BTR70, BMP2, T72, and several variants: three variants

TABLE VI
CONFUSION MATRIXES OF DIFFERENT METHODS
ON THREE TARGETS DATASET

(a) Confusion matrix using LC-KSVD				
Target	T72	BTR70	BMP2	Avg
T72(S7)	90.58%	1.05%	8.38%	93.26%
T72(S812)	91.28%	1.54%	7.18%	
T72(S132)	96.43%	0.00%	3.57%	
BTR70	1.02%	96.43%	2.55%	
BMP2(SC21)	1.53%	1.02%	97.45%	
BMP2(S9566)	5.10%	5.10%	89.80%	
BMP2(S9563)	6.67%	2.56%	90.77%	
(b) Confusion matrix using LLC				
Target	T72	BTR70	BMP2	Avg
T72(S7)	90.05%	4.71%	5.24%	91.87%
T72(S812)	90.26%	4.10%	5.64%	
T72(S132)	95.92%	1.02%	3.06%	
BTR70	4.08%	94.39%	1.53%	
BMP2(SC21)	3.57%	1.02%	95.41%	
BMP2(S9566)	8.67%	2.04%	89.29%	
BMP2(S9563)	6.67%	5.64%	87.69%	
(c) Confusion matrix using LcLcDDL				
Target	T72	BTR70	BMP2	Avg
T72(S7)	96.86%	1.57%	1.57%	96.41%
T72(S812)	94.87%	3.59%	1.54%	
T72(S132)	96.94%	0.00%	3.06%	
BTR70	2.04%	97.96%	0.00%	
BMP2(SC21)	3.06%	0.51%	96.43%	
BMP2(S9566)	3.57%	1.02%	95.41%	
BMP2(S9563)	3.59%	0.00%	96.41%	

of BMP2 (sn-9596, sn-9566, and sn-c21), three variants of T72 (sn-132, sn-812, and sn-s7) are used together to evaluate the performance of the proposed recognition method and the ability to deal with configuration variations. Only BTR70 (sn-c71), BMP2 (sn-c21), and T72 (sn-132) are included in the train set with depression angle 17°, and all targets including the variants construct the test set with depression angle 15°. Details of the train and test samples of the three targets recognition are listed in Table IV.

Detailed classification accuracy of the proposed LcLcDDL method and other comparison methods: SRC, K-SVD, LC-KSVD, and LLC are listed in Table V. To further compare the performance of the LcLcDDL method with LC-KSVD and LLC methods, the confusion matrixes of these three methods are given in Table VI. All these methods are used in the feature coding stage as explained in Section II. As discussed in Section V-D, Fig.6(a) shows that when the dictionary size is 2048, the classification accuracy of LcLcDDL is the better than that of 1024 by 0.37%, but the training computational burden and the storage space is about twice than that of 1024. Meanwhile, Fig.6(b) shows that when the sparsity is 15, the performance of LcLcDDL is the highest. Comprehensive consideration these factors, the dictionary size $N = 1024$, and sparsity $T = 15$ are adopted for these methods. Other parameters σ , β , and λ are selected as described in Section V-D.

TABLE VII
TARGETS TYPE AND NUMBER OF TRAINING AND TEST SAMPLES FOR TEN TARGETS RECOGNITION

Targets	T72			BRT70	BMP2			BRT60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
	132	S7	812		C21	9566	9563							
Train 17°	232	0	0	233	233	0	0	256	299	298	299	299	299	299
Test 15°	196	191	195	196	196	196	195	195	274	274	274	273	274	274

TABLE VIII
DETAILED ACCURACIES OF DIFFERENT METHODS ON TEN TARGETS DATASET

Methods	T72	BTR70	BMP2	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234	Avg
SRC	77.49%	83.16%	78.36%	83.59%	84.67%	84.31%	86.50%	81.68%	85.77%	85.77%	82.11%
K-SVD	85.57%	86.22%	83.30%	86.67%	82.85%	83.58%	86.50%	82.78%	81.39%	86.50%	84.42%
LC-KSVD	86.08%	88.27%	86.54%	87.18%	87.59%	87.23%	87.59%	85.71%	89.42%	88.69%	87.20%
LLC	86.08%	86.73%	84.50%	86.67%	85.40%	85.77%	83.21%	84.98%	85.77%	90.15%	85.76%
LcLcDDL	91.07%	91.33%	89.78%	90.26%	92.34%	89.42%	92.34%	90.11%	90.15%	95.26%	91.07%

As shown in Table V and Table VI, the classification accuracy of the proposed recognition method is higher than the remains. As for the variants, when compared with Table VI(a) and Table VI(b), there is smaller change of P_{cc} among the T72s and BMP2s in Table VI(c). This result indicates that the LcLcDDL recognition method is more robust to inner-class variability. These results demonstrate that the proposed recognition method is effective and can also deal with the configuration variation.

2) *Ten Targets Recognition Results and Analysis*: In this experiment, both ten distinct types of targets and variants are used in this ten targets recognition experiment. Ten targets: BTR70 (sn-c71), BMP2 (sn-c21), T72 (sn-132), BRDM2, BTR60, T62, 2S1, ZSU234, ZIL131 and D7 are used as the train set with depression angle 17°, and fourteen serials BTR70 (sn-c71), BMP2 (sn-9596, sn-9566, and sn-c21), T72 (sn-132, sn-812, and sn-s7), BRDM2, BTR60, T62, 2S1, ZSU234, ZIL131 and D7 construct the test set with depression angle 15°. Details of the train and test samples of the ten targets recognition are listed in Table VII.

As in three targets recognition, the methods: SRC, K-SVD, LC-KSVD, and LLC are also used as comparison in here. Detailed classification accuracy of the proposed LcLcDDL method and comparison methods are shown in Table VIII, the confusion matrixes of LC-KSVD, LLC and the proposed LcLcDDL method are given in Table IX, Table X and Table XI respectively. The dictionary size $N = 2048$, and sparsity $T = 15$ are adopted for these methods in ten targets recognition. Other parameters σ , β , and λ are selected as described in Section V-D.

As shown in Table VIII, Table IX, Table X and Table XI, the classification accuracy of the proposed recognition method is higher than others. These results demonstrate the effectiveness and robustness of the proposed method again, no matter the target with or without variants.

3) *Outlier Rejection Performance*: The above experiments are conducted under Extended Operating Conditions

(EOCs) [42], the configurations and depressions are different between the training set and test set (T72s and BMP2s, 17° in training set vs. 15° in testing set). But, the possible vehicle targets encountered in real world application can have an enormous number of targets that are not in the training set. Thus, the outlier rejection needs to be considered.

The outlier includes the confusers (i.e. targets not in the training set, such as civil vehicle) and the clutters (natural clutter or artificial clutter). Same as [43], D7 and 2S1 are treated as the confusers. As for the clutter, the clutter chips are generated from the clutter scene SAR images, including trees, buildings, streetlight, etc, are employed (clutter chips are random selected from the clutter set with size of 93×93 ($18.6m \times 18.6m$), the number of clutter chips is 548, same to the sum of D7 and 2S1). Eight targets: BTR70 (sn-c71), BMP2 (sn-c21), T72 (sn-132), BRDM2, BTR60, T62, ZSU234, and ZIL131 are used as the train set, and BTR70 (sn-c71), BMP2 (sn-9596, sn-9566, and sn-c21), T72 (sn-132, sn-812, and sn-s7), BRDM2, BTR60, T62, ZSU234, ZIL131, confusers (2S1, D7) and clutters construct the test set. Details number of the train and test samples can be found in Table VII.

As described in Section II, the classifier employed in this paper is LSVM, and Platt [44] proposes a method to make SVM output posterior probability.¹ Thus, we can refuse outliers according to the posterior probability. If the posterior probability of a sample is bigger than a specific threshold T , this sample is trusted as target, and a sample is judged as outlier if its posterior probability is lower than T .

In this paper, we evaluate the outlier rejection performance of the proposed method and other comparison methods from two aspects. By changing the threshold T , the receiver operating characteristic (ROC) curves for P_d versus P_{fa} can be obtained. ROC curve is one of the most widely used

¹In this paper, we use the LIBSVM tools downloaded from <http://www.csie.ntu.edu.tw/~cjlin/libsvm> to train the classifier. This tools also adopt John C. Platt's method to produce posterior probability. The only thing need to do to produce posterior probability is just predefining an optional parameter.

TABLE IX
CONFUSION MATRIX OF LLC ON TEN TARGETS DATASET

Targets	T72	BTR70	BMP2	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
T72(S7)	80.63%	0.52%	2.09%	0.00%	2.62%	0.00%	0.00%	8.38%	2.09%	3.66%
T72(S812)	85.64%	0.51%	2.05%	0.51%	3.59%	0.00%	0.00%	5.13%	1.03%	1.54%
T72(S132)	91.84%	0.00%	1.53%	0.00%	1.02%	0.00%	0.51%	3.06%	1.02%	1.02%
BTR70	1.02%	88.27%	2.55%	4.08%	3.06%	1.02%	0.00%	0.00%	0.00%	0.00%
BMP2(SC21)	1.53%	2.55%	93.37%	0.51%	0.00%	1.53%	0.00%	0.00%	0.51%	0.00%
BMP2(S9566)	3.06%	4.08%	84.69%	3.06%	3.06%	0.51%	0.00%	1.02%	0.00%	0.51%
BMP2(S9563)	2.05%	4.10%	81.54%	1.54%	3.08%	0.00%	2.05%	1.03%	2.05%	2.56%
BTR60	2.05%	1.03%	0.51%	87.18%	0.00%	1.54%	4.10%	1.03%	0.00%	2.56%
2S1	1.46%	1.09%	2.92%	1.09%	87.59%	1.09%	1.82%	1.82%	1.09%	0.00%
BRDM2	1.82%	1.46%	1.09%	1.09%	2.55%	87.23%	1.09%	1.82%	1.09%	0.73%
D7	1.82%	0.36%	1.82%	1.09%	1.46%	1.09%	87.59%	1.46%	1.82%	1.46%
T62	2.93%	0.73%	0.73%	0.73%	2.20%	1.83%	3.30%	85.71%	1.10%	0.73%
ZIL131	2.55%	0.73%	2.19%	0.73%	1.09%	1.46%	1.09%	0.73%	89.42%	0.00%
ZSU234	2.19%	0.73%	1.09%	1.09%	1.09%	1.82%	1.82%	0.73%	0.73%	88.69%

TABLE X
CONFUSION MATRIX OF LC-KSVD ON TEN TARGETS DATASET

Targets	T72	BTR70	BMP2	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
T72(S7)	80.63%	0.52%	2.09%	0.00%	2.62%	0.00%	0.00%	8.38%	2.09%	3.66%
T72(S812)	85.64%	0.51%	2.05%	0.51%	3.59%	0.00%	0.00%	5.13%	1.03%	1.54%
T72(S132)	91.84%	0.00%	1.53%	0.00%	1.02%	0.00%	0.51%	3.06%	1.02%	1.02%
BTR70	1.02%	88.27%	2.55%	4.08%	3.06%	1.02%	0.00%	0.00%	0.00%	0.00%
BMP2(SC21)	1.53%	2.55%	93.37%	0.51%	0.00%	1.53%	0.00%	0.00%	0.51%	0.00%
BMP2(S9566)	3.06%	4.08%	84.69%	3.06%	3.06%	0.51%	0.00%	1.02%	0.00%	0.51%
BMP2(S9563)	2.05%	4.10%	81.54%	1.54%	3.08%	0.00%	2.05%	1.03%	2.05%	2.56%
BTR60	2.05%	1.03%	0.51%	87.18%	0.00%	1.54%	4.10%	1.03%	0.00%	2.56%
2S1	1.46%	1.09%	2.92%	1.09%	87.59%	1.09%	1.82%	1.82%	1.09%	0.00%
BRDM2	1.82%	1.46%	1.09%	1.09%	2.55%	87.23%	1.09%	1.82%	1.09%	0.73%
D7	1.82%	0.36%	1.82%	1.09%	1.46%	1.09%	87.59%	1.46%	1.82%	1.46%
T62	2.93%	0.73%	0.73%	0.73%	2.20%	1.83%	3.30%	85.71%	1.10%	0.73%
ZIL131	2.55%	0.73%	2.19%	0.73%	1.09%	1.46%	1.09%	0.73%	89.42%	0.00%
ZSU234	2.19%	0.73%	1.09%	1.09%	1.09%	1.82%	1.82%	0.73%	0.73%	88.69%

approaches to assess the performance of outlier rejection. Here, the probability of detection P_d is defined as the percentage of targets that are correctly detection, and the probability of false alarm P_{fa} is defined as the percentage of outliers that are misjudged as targets. To further evaluate the performance of the proposed method and other comparison methods in a consistent manner, we select $P_d = 90\%$ as the operating point to compare the recognition accuracy of these methods (the same as [45]). The parameters used in these two experiments are the same as ten targets recognition.

The ROC curves of the proposed LcLcDDL method and other comparison methods are shown in Fig.7. Fig.7(a) is ROC curves for confuser rejection and Fig.7(b) is ROC curves for clutter rejection. As shown in Fig.7, the proposed method

achieves better performance than other comparison methods in both confuser rejection and clutter rejection. Specially, in Fig.7(a), when $P_d = 90\%$, P_{fa} of the proposed method is much lower than that of other comparison methods, the similar result can be observed in Fig.7(b).

When the operating point ($P_d = 90\%$ in here) is defined, the corresponding threshold T can be obtained. The recognition accuracy of LcLcDDL method and other methods are tabulated in Table XII. From Table XII, we can see that the recognition accuracy is higher than other comparison methods, and the outlier rejection performance is also better than other methods.

The experimental results shown in Fig.7 and Table XII demonstrate that the proposed LcLcDDL method is available

TABLE XI
CONFUSION MATRIX OF LcLcDDL ON TEN TARGETS DATASET

Targets	T72	BTR70	BMP2	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
T72(S7)	89.53%	0.00%	1.05%	4.71%	3.14%	0.00%	0.00%	1.57%	0.00%	0.00%
T72(S812)	88.72%	1.54%	0.00%	5.64%	2.56%	0.00%	0.00%	1.54%	0.00%	0.00%
T72(S132)	94.90%	0.00%	0.00%	3.06%	0.00%	1.02%	0.00%	1.02%	0.00%	0.00%
BTR70	0.00%	91.33%	0.00%	7.14%	1.53%	0.00%	0.00%	0.00%	0.00%	0.00%
BMP2(SC21)	0.00%	0.51%	93.37%	4.59%	0.51%	1.02%	0.00%	0.00%	0.00%	0.00%
BMP2(S9566)	2.55%	2.04%	88.27%	5.61%	0.51%	1.02%	0.00%	0.00%	0.00%	0.00%
BMP2(S9563)	1.03%	1.03%	87.69%	9.23%	0.00%	0.00%	0.51%	0.00%	0.51%	0.00%
BTR60	0.51%	2.56%	1.03%	90.26%	2.05%	0.00%	1.03%	1.03%	0.51%	1.03%
2S1	0.73%	0.73%	1.46%	1.82%	92.34%	1.09%	0.36%	0.36%	0.73%	0.36%
BRDM2	0.73%	1.09%	2.19%	1.46%	1.09%	89.42%	1.46%	0.73%	1.09%	0.73%
D7	1.09%	1.09%	0.36%	0.36%	0.73%	1.09%	92.34%	0.73%	1.46%	0.73%
T62	2.56%	0.37%	1.83%	0.73%	1.47%	0.73%	0.00%	90.11%	1.83%	0.37%
ZIL131	1.82%	0.36%	1.09%	1.46%	1.09%	1.09%	0.73%	0.73%	90.15%	1.46%
ZSU234	0.73%	0.36%	0.73%	0.36%	0.73%	0.73%	0.36%	0.73%	0.00%	95.26%

TABLE XII
DETAILED ACCURACIES OF DIFFERENT METHODS ON EIGHT TARGETS DATASET

Methods	T72	BTR70	BMP2	BTR60	BRDM2	T62	ZIL131	ZSU234	Outlier	Avg
SRC	66.15%	83.67%	71.21%	85.64%	85.77%	83.15%	87.96%	86.50%	81.57%	79.13%
K-SVD	69.93%	81.63%	76.83%	89.23%	86.86%	85.35%	83.58%	88.32%	85.58%	81.90%
LC-KSVD	74.23%	84.18%	75.47%	89.23%	88.32%	86.81%	91.24%	89.78%	86.04%	83.50%
LLC	74.40%	80.10%	73.42%	88.21%	88.32%	86.81%	86.13%	88.69%	86.68%	82.67%
LcLcDDL	77.49%	91.33%	73.76%	93.33%	91.24%	90.84%	92.34%	96.35%	89.96%	86.54%

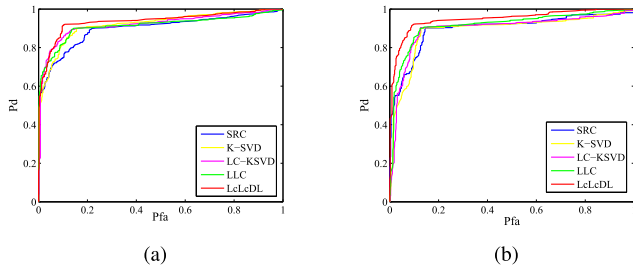


Fig. 7. Experimental results of outlier rejection. (a) ROC curves for the confuser rejection. (b) ROC curves for the clutter rejection.

to reject outliers as well as to accurately recognize the trusted targets. What is more, the performance of the proposed method is also better than other comparison methods.

VI. CONCLUSION

In this paper, we propose a novel SAR recognition method based on the ASCM and discriminate dictionary learning. In the low-level local features extraction stage, the ASCM is utilized to convolve with the SAR patches, thus the low-level local features contain not only the local information of the patches, but also the physical information of the targets. Meanwhile, GA is adopted to select the most discriminative combinations of the ASCM parameters. To further improve the discriminative power of the low-level local feature, a discriminate dictionary learning method LcLcDDL incorporating

the label information and local geometric information into the sparse representation is proposed in the feature coding stage. When the dictionary is learned, the sparse coefficients of the low-level local feature are obtained by sparse representation. In the feature pooling stage, SPM model is used to generate the high-level global feature each SAR patch. Experimental results based on the MSTAR dataset demonstrate the effectiveness of the low-level local feature and the LcLcDDL method, the results also show that the proposed recognition method has a better performance than other methods on real SAR images. Nevertheless, constructing deep multi-level descriptions about SAR images may further improve the target recognition performance, especially when these descriptions imply physical semantic, which is also the focus of our future work.

REFERENCES

- [1] D. E. Dudgeon and R. T. Lacoss, "An overview of automatic target recognition," *Lincoln Lab. J.*, vol. 6, no. 1, pp. 3–10, 1993.
- [2] L. M. Novak, G. J. Owirka, and C. M. Netishen, "Performance of a high-resolution polarimetric SAR automatic target recognition system," *Lincoln Lab. J.*, vol. 6, no. 1, pp. 11–23, 1993.
- [3] L. M. Novak, G. J. Owirka, W. S. Brower, and A. L. Weaver, "The automatic target-recognition system in SAIP," *Lincoln Lab. J.*, vol. 10, no. 2, pp. 187–202, 1997.
- [4] J. A. O'Sullivan, M. D. DeVore, V. Kedia, and M. I. Miller, "SAR ATR performance using a conditionally Gaussian model," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 1, pp. 91–108, Jan. 2001.
- [5] M. D. DeVore and J. A. O'Sullivan, "Performance complexity study of several approaches to automatic target recognition from SAR images," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 2, pp. 632–648, Apr. 2002.

- [6] Y. Chen, E. Blasch, T. Qian, and E. Blasch, "Experimental feature-based SAR ATR performance evaluation under different operational conditions," *Proc. SPIE*, vol. 6968, pp. 69680F-1–69680F-12, May 2008.
- [7] C. Clemente, L. Pallotta, D. Gaglione, A. De Maio, and J. J. Soraghan, "Automatic target recognition of military vehicles with Krawtchouk moments," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 1, pp. 493–500, Feb. 2017.
- [8] M. Amoon and G.-A. Rezai-Rad, "Automatic target recognition of synthetic aperture radar (SAR) images based on optimal selection of Zernike moments features," *IET Comput. Vis.*, vol. 8, no. 2, pp. 77–85, Apr. 2014.
- [9] S. Papson and R. M. Narayanan, "Classification via the shadow region in SAR imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 2, pp. 969–980, Apr. 2012.
- [10] J.-I. Park, S.-H. Park, and K.-T. Kim, "New discrimination features for SAR automatic target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 3, pp. 476–480, May 2013.
- [11] S. K. Rogers et al., "Neural networks for automatic target recognition," *Neural Netw.*, vol. 8, no. 7, pp. 1153–1184, 1995.
- [12] S. Chen, H. Wang, F. Xu, and Y. Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.
- [13] J. Ding, B. Chen, H. Liu, and M. Huang, "Convolutional neural network with data augmentation for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 364–368, Mar. 2016.
- [14] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [15] J. Martens, "Deep learning via Hessian-free optimization," in *Proc. Int. Conf. Mach. Learn.*, 2010, pp. 735–742.
- [16] J. Wright, Y. Ma, J. Mairal, G. Sapiro, T. S. Huang, and S. Yan, "Sparse representation for computer vision and pattern recognition," *Proc. IEEE*, vol. 98, no. 6, pp. 1031–1044, Jun. 2010.
- [17] J. Wright, A. Y. Yang, A. Ganesh, S. S. Sastry, and Y. Ma, "Robust face recognition via sparse representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 2, pp. 210–227, Feb. 2009.
- [18] J. Yang, K. Yu, Y. Gong, and T. Huang, "Linear spatial pyramid matching using sparse coding for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, Jun. 2009, pp. 1794–1801.
- [19] J. J. Thiagarajan, K. N. Ramamurthy, P. Knee, A. Spanias, and V. Berisha, "Sparse representations for automatic target classification in SAR images," in *Proc. 4th Int. Symp. Commun., Control Signal Process. (ISCCSP)*, Limassol, Cyprus, 2010, pp. 1–4.
- [20] G. Dong, G. Kuang, N. Wang, L. Zhao, and J. Lu, "SAR target recognition via joint sparse representation of monogenic signal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3316–3328, Jul. 2015.
- [21] L. Zhang, L. Sun, B. Zou, and W. M. Moon, "Fully polarimetric SAR image classification via sparse representation and polarimetric features," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 8, pp. 3923–3932, Aug. 2015.
- [22] F. Yang, W. Gao, B. Xu, and J. Yang, "Multi-frequency polarimetric SAR classification based on riemannian manifold and simultaneous sparse representation," *Remote Sens.*, vol. 7, no. 7, pp. 8469–8488, 2015.
- [23] X. Sun, N. M. Nasrabadi, and T. D. Tran, "Task-driven dictionary learning for hyperspectral image classification with structured sparsity constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4457–4471, Aug. 2015.
- [24] M. Aharon, M. Elad, and A. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, Nov. 2006.
- [25] J. Mairal, F. Bach, J. Ponce, and G. Sapiro, "Online dictionary learning for sparse coding," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Montreal, QC, Canada, 2009, pp. 689–696.
- [26] Q. Zhang and B. Li, "Discriminative K-SVD for dictionary learning in face recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, San Francisco, CA, USA, Jun. 2010, pp. 2691–2698.
- [27] Z. Jiang, Z. Lin, and L. S. Davis, "Label consistent K-SVD: Learning a discriminative dictionary for recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2651–2664, Nov. 2013.
- [28] K. Yu, T. Zhang, and Y. Gong, "Nonlinear learning using local coordinate coding," in *Proc. Conf. Workshop Neural Inf. Process. Syst.*, 2009, pp. 2223–2231.
- [29] L. C. Potter and R. L. Moses, "Attributed scattering centers for SAR ATR," *IEEE Trans. Image Process.*, vol. 6, no. 1, pp. 79–91, Jan. 1997.
- [30] J. B. Keller, "Geometrical theory of diffraction," *J. Opt. Soc. Amer.*, vol. 52, no. 2, pp. 116–130, 1962.
- [31] H. Liu, B. Jiu, F. Li, and Y. Wang, "Attributed scattering center extraction algorithm based on sparse representation with dictionary refinement," *IEEE Trans. Antennas Propag.*, vol. 65, no. 5, pp. 2604–2614, May 2017.
- [32] W. L. Price, "Global optimization by controlled random search," *J. Optim. Theory Appl.*, vol. 40, no. 3, pp. 333–348, 1983.
- [33] M. Stein, "Large sample properties of simulations using Latin hypercube sampling," *Technometrics*, vol. 29, no. 2, pp. 143–151, 1987.
- [34] J. E. Baker, "Reducing bias and inefficiency in the selection algorithm," in *Proc. 2nd Int. Conf. Genetic Algorithms*, 1987, pp. 14–21.
- [35] C. M. Bishop, *Pattern Recognition and Machine Learning* (Information Science and Statistics). New York, NY, USA: Springer-Verlag, 2006.
- [36] J. Wang, J. Yang, K. Yu, F. Lv, T. Huang, and Y. Gong, "Locality-constrained Linear coding for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, San Francisco, CA, USA, Jun. 2010, pp. 3360–3367.
- [37] G. Golub, P. C. Hansen, and D. P. O'leary, "Tikhonov regularization and total least squares," *SIAM J. Matrix Anal. Appl.*, vol. 21, no. 1, pp. 185–194, 1999.
- [38] K. Grauman and T. Darrell, "The pyramid match kernel: Discriminative classification with sets of image features," in *Proc. 10th IEEE Int. Conf. Comput. Vis. (ICCV)*, Beijing, China, vol. 2, Oct. 2005, pp. 1458–1465.
- [39] Z. Jianxiong, S. Zhiguang, C. Xiao, and F. Qiang, "Automatic target recognition of SAR images based on global scattering center model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3713–3729, Oct. 2011.
- [40] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 20, no. 2, pp. 91–110, 2004.
- [41] H. Lee, A. Battle, R. Raina, and A. Y. Ng, "Efficient sparse coding algorithms," in *Proc. Conf. Adv. Neural Inf. Process. Syst.*, vol. 19, 2007, pp. 801–808.
- [42] E. Keydel, S. Lee, and J. Moore, "MSTAR extended operating conditions: A tutorial," *Proc. SPIE*, vol. 2757, pp. 228–242, Jun. 1996.
- [43] Y. Sun, Z. Liu, S. Todorovic, and J. Li, "Adaptive boosting for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 43, no. 1, pp. 112–125, 2007.
- [44] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Adv. Large Margin Classifiers*, 2000, pp. 1–12.
- [45] L. M. Novak, S. D. Halversen, G. Owirka, and M. Hiett, "Effects of polarization and resolution on SAR ATR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 1, pp. 102–116, Jan. 1997.



Tingli Li received the B.S. degree from the School of Electronic and Information Engineering, Henan Polytechnic University, Jiaozuo, China, in 2012. He is currently pursuing the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing, Xidian University.

His main research interests include synthetic aperture radar (SAR) automatic target recognition, SAR image processing, and pattern recognition.



Lan Du (M'11–SM'17) received the B.S., M.S., and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 2001, 2004, and 2007, respectively.

From 2007 to 2009, she was a Post-Doctoral Research Associate with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA, where she was involved in research. She is currently a Professor with the National Laboratory of Radar Signal Processing, Xidian University. Her research is supported by NSFC for Excellent Young Scholars and Chang Jiang Scholars Program for Young Scholars. Her main research interests are in the fields of statistical signal processing and machine learning with application to radar target recognition. She was a recipient of the National Excellent Doctoral Dissertation of China for her Ph.D. dissertation in 2009.